

SIMATS ENGINEERING
ASSESSMENT TOOL 2 : Scenario-Based

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)*

Assessment Tool Weightage: 20%

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I Harshika (192573008) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Harshika

Assessment Tasks

Scenario-Based Questions

1. A media company wants to migrate its video streaming platform to the cloud.
The system must handle millions of concurrent users with minimal buffering.
Design a cloud architecture using scalability and caching techniques.
Explain how load balancing and auto-scaling can ensure smooth streaming.
Discuss cost optimization strategies for such high-demand workloads.
2. A fintech startup is deploying applications using containers across multiple cloud providers.
They face challenges in maintaining consistency and managing deployments.
Propose a solution using container orchestration and multi-cloud strategies.
Explain how service discovery and scaling can be handled efficiently.
Highlight the benefits and risks of a multi-cloud deployment model.
3. An e-learning platform experiences data loss due to accidental deletion in the cloud.
The organization needs a robust backup and disaster recovery strategy.
Design a solution using cloud-native backup, replication, and versioning.

Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved.
Provide steps to ensure business continuity during failures.

4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data.
Some applications must remain on-premises due to compliance requirements.
Design a hybrid architecture that ensures secure communication between environments.
Explain how identity management and encryption should be implemented.
Discuss challenges in managing hybrid cloud environments.
5. A SaaS provider needs to ensure high availability and fault tolerance for its application.
The system must continue operating even if one region goes down.
Propose a multi-region deployment strategy in the cloud.
Explain how data replication and failover mechanisms should be configured.
Discuss monitoring and alerting strategies for proactive issue detection.

Scenario-Based Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Scenario	20%	Demonstrates complete understanding of the scenario context	Good understanding with minor gaps	Partial understanding	Fails to understand the scenario
Identification of Risks / Issues	20%	Identifies all major issues correctly	Identifies most issues	Limited issue identification	Misses major issues
Application of Concepts	25%	Applies virtualization concepts effectively	Applies concepts with minor mistakes	Partially correct application	Weak application
Decision Justification	20%	Decisions are well justified and technically strong	Reasonable justification	Weak justification	No useful justification
Clarity	15%	Clear and well-organized response	Generally clear	Somewhat unclear	Poorly presented

1. Cloud Architecture for Video Streaming Platform.

A media company requires a cloud-based architecture that supports millions of concurrent users with minimal buffer. The solution should provide high availability, scalability and cost efficiency.

Cloud Architecture Design

- Deploy the application across multiple availability zones.
- Use a Content Delivery Network to cache video content close to users.
- Store videos in cloud object storage.
- Deploy application servers behind a load balancer.

Load Balancing and Auto Scaling

- The load balancer distributes incoming traffic evenly among application servers.
- Auto Scaling monitors CPU utilization and network traffic.
- During peak hours, additional virtual machines are created automatically.
- During low demand, unused resources are terminated to reduce costs.

Cost optimization

- Use reserved instances for predictable workload.
- Use spot instances for background processing.
- Enable automatic resource scaling.
- Compress and cache video files to reduce bandwidth cost.

2. Container Orchestration in Multi-Cloud.

A fintech startup deploys applications using containers across multiple cloud providers. managing consistency and deployments becomes challenging.

Proposed solution

- Use Kubernetes for container orchestration.
- Store container image in a centralized container registry.
- Use CI/CD Pipeline for automated deployment.

Service Discovery and Scaling.

- Kubernetes Service Discovery automatically locates running services.
- Horizontal Pod Autoscaler increases or decreases containers instances based on CPU and memory utilization.

Benefits :

- High availability
- Vendor Independence
- Improved scalability
- Better disaster recovery.

Risks:

- Increased management complexity
- Higher operational costs
- Security configuration challenges
- Network latency between cloud providers.

3. Backup and Disaster Recovery strategy.

An e-learning platform experienced accidental data deletion and requires a reliable backup and disaster recovery solution.

Solution Design

- Enable automatic cloud backups.
- Store backup copies in different geographic regions.
- Enable object versioning.
- Use database replication
- Schedule regular backup testing.

RTO and RPO

- Recovery Time objective (RTO): Reduce downtime by using automated failover and standby servers.
- Recovery Point objective (RPO): Perform frequent backups and continuous replication to minimize data loss.

Business Continuity:

- Maintain redundant infrastructure.
- Perform regular disaster recovery drills.
- monitor backup health.
- Document recovery Procedures.

4. Hybrid cloud Architecture.

A government agency stores sensitive and non-sensitive data using a hybrid cloud model.

Hybrid cloud Design:

- Sensitive applications remain on-premises.
- Public cloud hosts non-sensitive applications.
- Secure VPN or dedicated private connections link both environments.
- unified management platform monitors resources.

Identity Management:

- Implement single sign-on (SSO)
- Use multi-Factor Authentication (MFA)
- Apply Role-Based Access control (RBAC).

Encryption:

- Encrypt data during transmission using TLS.
- Encrypt stored using AES-256.
- Secure encryption keys using a key management service (KMS).

Challenges ÷

- Complex infrastructure management.
- Compliance monitoring
- Security Policy Consistency
- Integration b/w cloud and on-premises systems.

5. Multi-Region Deployment Strategy

A SaaS provider requires continuous service availability even if an entire cloud region fails.

Multi-Region Deployment ÷

- Deploy applications in multiple geographic regions.
- Use global Load Balancers to route traffic.
- Configure active-active deployment for maximum availability.

Data Replication and Failover

- Enable synchronous replication within a region.
- Use asynchronous replication across regions.
- Configure automatic failover when a region becomes unavailable.

Monitoring and Alerting:-

- Monitor CPU, memory, network, and storage performance.
- configure real-time alerts for failures.
- Use centralized logging and dashboards.
- Perform regular health checks.

Conclusion:

Multi-region deployment with automated failover, application, replication and monitoring ensures high availability, fault tolerance and uninterrupted cloud services.